Mathematical Statistics

Interval Estimation

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Definition

- The inference in an interval estimation (set estimation) problem is the statement $\theta \in \mathcal{C}$, where $\mathcal{C} = \mathcal{C}(x)$ is a set determined by the value of the data $\boldsymbol{X} = \boldsymbol{x}$.
- An interval estimate of a real-valued parameter $\theta$is any pair of functions, $L(\boldsymbol{x})$ and $U(\boldsymbol{x})$, of a sample that satisfy $L(\boldsymbol{x}) \leq U(\boldsymbol{x})$. If $\boldsymbol{X} = \boldsymbol{x}$ is observed, the inference $L(\boldsymbol{x}) \leq \theta \leq U(\boldsymbol{x})$ is made.
- The random interval $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ is called an interval estimator.

Definition - Coverage Probability

- For an interval estimator $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ of a parameter θ, the coverage probability of $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ is

$$P_\theta \left(\theta \in [L(\boldsymbol{X}), U(\boldsymbol{X})]\right).$$

- The confidence coefficient is

$$\inf_\theta P_\theta \left(\theta \in [L(\boldsymbol{X}), U(\boldsymbol{X})]\right).$$

- Interval estimators, together with a measure of confidence coefficient are sometime known as confidence intervals.
- A confidence interval with confidence coefficient equal to some value, say $1-\alpha$, is called $1-\alpha$ confidence interval.

Definition - Coverage Probability

- For an interval estimator $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ of a parameter θ, the coverage probability of $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ is

$$P_\theta \left(\theta \in [L(\boldsymbol{X}), U(\boldsymbol{X})]\right).$$

- The confidence coefficient is

$$\inf_\theta P_\theta \left(\theta \in [L(\boldsymbol{X}), U(\boldsymbol{X})]\right).$$

- Interval estimators, together with a measure of confidence coefficient are sometime known as confidence intervals.
- A confidence interval with confidence coefficient equal to some value, say $1-\alpha$, is called $1-\alpha$ confidence interval.

Example 9.1.6 on uniform interval estimator

- $X_1, \ldots, X_n \sim \mathit{uniform}(0, \theta)$, where $0 < \theta < \infty$, and let $Y = X_{(n)}$.
- Find the coverage probability for the two interval estimators, $[aY, bY]$, $1 \leq a < b$, and $[Y + c, Y + d]$, $0 \leq c < d$.
- Solution: note that the density of Y is: $f_Y(y) = ny^{n-1}/\theta^n$, $0 \leq y \leq \theta$, and the pdf of $T = Y/\theta$ is $f_T(t) = nt^{n-1}, 0 \leq t \leq 1$. Thus,

$$P_\theta(\theta \in [aY, bY]) = P_\theta(\frac{1}{b} \leq T \leq \frac{1}{a}) =$$

$$(\frac{1}{a})^n - (\frac{1}{b})^n,$$

implying the confidence coefficient of the interval is $(\frac{1}{a})^n - (\frac{1}{b})^n$.

Example 9.1.6 on uniform interval estimator

- $$P_\theta(\theta \in [Y+c, Y+d]) = P_\theta(1-\frac{d}{\theta} \leq T \leq 1-\frac{c}{\theta}) =$$

  $$(1-\frac{c}{\theta})^n - (1-\frac{d}{\theta})^n.$$

  Since

  $$(1-\frac{c}{\theta})^n - (1-\frac{d}{\theta})^n \to 0 \text{ as } \theta \to \infty,$$

  the confidence coefficient of the second interval estimator is 0.

Four Construction Methods

- Probability Inequalities
- Inverting a Test
- Pivots
- Large Sample Approximations

NOTE: Optimal confidence intervals are referred as confidence intervals that are as short as possible but we will not discuss optimality.

Probability Inequality Methods

Hoeffding's Inequality: for independent random variables $X_1, \ldots, X_n$ with strictly bounded by the intervals $[a_i, b_i]$, then

$$P(\mid \bar{X} - E(\bar{X}) \mid \geq t) \leq 2\exp\left(-\frac{2n^2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

- Let $X_1, \ldots, X_n \sim Bernoulli(p)$. By Hoeffding's Inequality, we have

  $$P(\mid \hat{p} - p \mid > \epsilon_n) \leq 2e^{-2n\epsilon_n^2}.$$

  Let

  $$\epsilon_n = \sqrt{\log(2/\alpha)/2n}.$$

  Then,

  $$P(\mid \hat{p} - p \mid > \epsilon_n) \leq \alpha.$$

- Therefore, $P(p \in C) \geq 1 - \alpha$, where $C = (\hat{p} - \epsilon_n, \hat{p} + \epsilon_n)$.

Inverting a Test

The theorem between acceptance region of tests and confidence sets:

- For each $\theta_0 \in \Theta$, let $A(\theta_0)$ be the acceptance region of a level α test of $H_0 : \theta = \theta_0$.
- For each $x \in \chi$, define a set $C(x)$ in the parameter space as follows:
$$C(x) = \{\theta_0 : x \in A(\theta_0)\}.$$
- Then the random set $C(X)$ is a $1 - \alpha$ confidence set. That is, for each θ,
$$P_\theta(\theta \in C(\boldsymbol{X})) = 1 - \alpha.$$
- Conversely, let $C(X)$ be a $1 - \alpha$ confidence set. For every $\theta_0 \in \Theta$, then $A(\theta_0) = \{x : \theta_0 \in C(x)\}$ is the acceptance region of a level α test of $H_0 : \theta = \theta_0$.

Proof of the Theorem

- For the first part, since $A(\theta_0)$ is the acceptance region of a level α test, $P_{\theta_0}(X \notin A(\theta_0)) \leq \alpha$ and therefore, $P_{\theta_0}(X \in A(\theta_0)) \geq 1 - \alpha$. Since θ_0 is arbitrary, we can write θ instead of θ_0. The above inequality, together with the definition of $C(X)$, shows that the coverage probability of the set $C(X)$ is given by

$$P_\theta(\theta \in C(X)) = P_\theta(X \in A(\theta)) \geq 1 - \alpha.$$

- For the second part, the Type I error probability for the test of $H_0 : \theta = \theta_0$ with acceptance region $A(\theta_0)$ is

$$P_{\theta_0}(X \notin A(\theta_0)) = P_{\theta_0}(\theta_0 \notin C(X)) \leq \alpha.$$

  So this is a level α test.

Example

- Suppose we used the LRT. We reject H_0 when

$$L(\theta_0)/L(\widehat{\theta}) \leq c.$$

- So, $C = \{\theta : L(\theta)/L(\widehat{\theta}) \geq c\}$.
- Let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$ with σ known.
- Then, the LRT of $H_0 : \mu = \mu_0$ rejects when

$$| \bar{X} - \mu_0 | \geq \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2}.$$

- Let $\boldsymbol{X} = (X_1, \ldots, X_n)$. Therefore, the acceptance region

$$A(\mu_0) = \left\{ \boldsymbol{X} : | \bar{X} - \mu_0 | < \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2} \right\}.$$

- Hence, the corresponding confidence interval is:

$$C(\boldsymbol{x}) = \left\{ \mu_0 : | \bar{X} - \mu_0 | \leq \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2} \right\}.$$

One-Sided Confidence Intervals

- The test inversion method is completely general in that we can invert any test and obtain a confidence set.
- The inversion of a two-sided test gave a two-sided interval. We can also obtain one-sided intervals by inverting one-sided tests.

One-Sided Interval Example

- Let $X_1, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$. We are interested in constructing a $1 - \alpha$ upper confidence interval for μ; that is, $C(\boldsymbol{x}) = (-\infty, U(\boldsymbol{x})]$.
- To obtain such an interval, we will invert one-sided tests of $H_0 : \mu = \mu_0$ versus $H_1 : \mu < \mu_0$. Note that we use the specification of H_1 to determine the form of the confidence interval here.
- The size α LRT of H_0 versus H_1 rejects H_0 if

$$\frac{\bar{X} - \mu_0}{S/\sqrt{n}} < -t_{n-1,\alpha},$$

where $S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$.

One-Sided Interval Example

- Thus, the acceptance region for this test is:

$$A(\mu_0) = \left\{ \boldsymbol{x} : \bar{x} \geq \mu_0 - t_{n-1,\alpha} \frac{s}{\sqrt{n}} \right\}.$$

- Thus, $1 - \alpha$ confidence set for μ is given as follows:

$$C(\boldsymbol{x}) = \left\{ \mu_0 : \bar{x} \geq \mu_0 - t_{n-1,\alpha} \frac{s}{\sqrt{n}} \right\}.$$

Pivots

- A function $Q(X_1, \ldots, X_n, \theta)$ is a pivot if the distribution of Q does not depend on θ. For example, if $X_1, \ldots, X_n \sim N(\theta, 1)$, then $\bar{X}_n - \theta \sim N(0, 1/n)$. So $Q = \bar{X}_n - \theta$ is a pivot.
- Let a and b be such that

$$P_\theta(a \leq Q(X, \theta) \leq b) = 1 - \alpha$$

for all θ. Then,

$$C(x) = \{\theta : a \leq Q(x, \theta) \leq b\}$$

has coverage $1 - \alpha$.

Example 1

Let $X_1, \ldots, X_n \sim N(\mu, \sigma)$.

- If σ is known, $Z = \sqrt{n}(\bar{X} - \mu)/\sigma \sim N(0, 1)$. We know that

$$P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}) = 1 - \alpha.$$

Hence,

$$C = \left(\bar{X} - \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2} \right).$$

- If σ is unknown,

$$C = \left(\bar{X} - \frac{S}{\sqrt{n}} t_{n-1,1-\alpha/2}, \bar{X} + \frac{S}{\sqrt{n}} t_{n-1,1-\alpha/2} \right),$$

where S is the sample variance, $S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$, because

$$T = \sqrt{n}(\bar{X} - \mu)/S \sim t_{n-1}.$$

Example 2

Let $X_1, \ldots, X_n \sim \mathit{Uniform}(0, \theta)$.

- Let $Q = X_{(n)}/\theta$. Then,

$$P(Q \leq t) = \prod_i P(X_i \leq t\theta) = t^n.$$

  So Q is a pivot. Let $c_n = \alpha^{1/n}$.
- Then,

$$P(Q \leq c_n) = \alpha.$$

Example 2, continued

- Since $P(Q \leq 1) = 1$,

$$\begin{aligned} 1 - \alpha &= P(c_n \leq Q \leq 1) \\ &= P\left(c_n \leq \frac{X_{(n)}}{\theta} \leq 1\right) \\ &= P\left(\frac{1}{c_n} \geq \frac{\theta}{X_{(n)}} \geq 1\right) \\ &= P\left(X_{(n)} \leq \theta \leq \frac{X_{(n)}}{c_n}\right). \end{aligned}$$

- So a $1 - \alpha$ confidence interval is

$$\left(X_{(n)}, \frac{X_{(n)}}{\alpha^{1/n}}\right).$$

Delta Method

- If $g'(a)$ exists and $n^b(X_n - a) \to X$ in distribution for $b > 0$, then

$$n^b(g(X_n) - g(a)) \to g'(a)X \text{ in distribution}$$

- The proof of the delta method uses Taylor's expansion and Slutsky's theorem, Since $X_n - a \to 0$ in probability, Taylor expansion:
$n^b(g(X_n) - g(a)) = n^b(X_n - a)(g'(a) + o_p(1))$.
- Using Slutsky's theorem and the fact that $n^b(X_n - a) \to X$ in distribution, we obtain the desired result.

Large Sample Confidence Intervals for $\tau(\theta)$

- Let $\widehat{\theta}$ be the MLE of θ. Under regularity conditions, we have that

$$\frac{\widehat{\theta}_n - \theta}{se} \xrightarrow{d} N(0,1),$$

where $\widehat{\theta}_n$ is the MLE and $se = 1/\sqrt{\mathcal{I}_n(\widehat{\theta})}$. So, this is an asymptotic pivot.

- An approximate confidence interval (Wald type) is

$$\left(\widehat{\theta}_n - z_{1-\alpha/2}se,\ \widehat{\theta}_n + z_{1-\alpha/2}se\right).$$

- By the Delta method, a confidence interval for $\pi(\theta)$ is

$$\left(\pi(\widehat{\theta}_n) - z_{1-\alpha/2}se|\pi'(\widehat{\theta})|,\ \pi(\widehat{\theta}_n) + z_{1-\alpha/2}se|\pi'(\widehat{\theta})|\right).$$

Example 10.4.1

- Let $X_1, \ldots, X_n \sim Bernoulli(p)$, and we want to construct a confidence interval for $p/(1-p)$.
- The MLE for p is $\widehat{p} = \sum_i X_i/n$ and $Var(\widehat{p}) = p(1-p)/n$.
- Delta method leads to $Var(\frac{\widehat{p}}{1-\widehat{p}}) \approx \frac{p(1-p)}{n}(\frac{1}{(1-p)^2}) = \frac{p}{n(1-p)^3}$.
- The $1-\alpha$ confidence interval is:

$$\frac{\widehat{p}}{1-\widehat{p}} - z_{1-\alpha/2}\sqrt{\widehat{Var}(\frac{\widehat{p}}{1-\widehat{p}})} \leq \frac{p}{1-p} \leq \sqrt{\frac{\widehat{p}}{1-\widehat{p}} + z_{1-\alpha/2}\widehat{Var}(\frac{}{1-}$$

Intervals based on Score statistic

- When applicable, the more restrictive form of the likelihood approximation, based on the score statistic, $S(X \mid \theta) = \frac{\partial}{\partial \theta} logL(\theta \mid X)$, can give better intervals.
- The random quantity

$$Q(X \mid \theta) = \frac{S(X \mid \theta)}{\sqrt{-E_\theta(\frac{\partial^2}{\partial \theta^2} l(\theta \mid X))}}$$

  has a $N(0,1)$ asymptotically, asymptotic pivital statistic.
- Thus, the set

$$\{\theta :\mid Q(x \mid \theta) \mid \leq z_{\alpha/2}\}$$

  is an approximate $1-\alpha$ confidence set,
- Wilks (1938) proved that these intervals have an asymptotically optimality property, the shortest in a certain class of intervals.

Example 10.4.2

- Let $X_1, \ldots, X_n \sim Bernoulli(p)$, and we want to construct a confidence interval for $p/(1-p)$.
- The MLE for p is $\widehat{p} = \sum_i X_i/n$. Let. $Y = \sum_{i=1}^{n} X_i$.
- The log-likelihood $l(\theta \mid Y) = log(\binom{n}{y}) + ylog(p) - (n-y)log(1-p)$, and the score statistic, based on Y: $S(y \mid \theta) = \frac{y}{p} - \frac{n-y}{1-p}$ and $E_p(\frac{\partial^2}{\partial p^2} l(p \mid Y)) = \frac{n}{p(1-p)}$
- Thus, $Q(Y \mid p) = \frac{\widehat{p}-p}{\sqrt{p(1-p)/n}}$
- An approximate $1-\alpha$ confidence interval for p is:

$$\{p :\mid \frac{\widehat{p}-p}{\sqrt{p(1-p)/n}} \mid \leq z_{\alpha/2}\}$$

The Likelihood Ratio Statistic Based Confidence Set

- We can also invert the asymptotic LRT to derive an confidence set. Note that we can test $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ by inverting the LRT, where θ is a $k \times 1$ vector.
- We do not reject H_0 if

$$-2 \log \left(\frac{L(\theta_0)}{L(\widehat{\theta})} \right) \leq \chi^2_{k,1-\alpha}.$$

  Hence, the set of non-rejected nulls is

$$C_n = \left\{ \theta : \frac{L(\theta)}{L(\widehat{\theta})} > e^{-\chi^2_{k,1-\alpha}/2} \right\}.$$

- It can be shown that $P_\theta(\theta \in C_n) \to 1 - \alpha$ for each θ.

Example

Let $X_1, \ldots, X_n \sim \mathit{Bernoulli}(p)$.

- Using the Wald statistic

$$\frac{\widehat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \xrightarrow{d} N(0,1).$$

- So, an approximate confidence interval is

$$\left(\hat{p} - z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}, \hat{p} + z_{1-\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\right).$$

- Using the LRT we obtain

$$C = \left\{ p : -2\log\left(\frac{p^Y(1-p)^{n-Y}}{\hat{p}^Y(1-\hat{p})^{n-Y}}\right) \leq \chi^2_{1,1-\alpha} \right\},$$

where $Y = \sum_{i=1}^n X_i$.

- These intervals are different but, for large n, they are nearly the same.

Methods for Evaluating Interval Estimators

- Lengths and coverage probability of an interval.
- For a given coverage probability, we like to find the confidence interval with the shortest length.

Example 9.3.1 (Optimizing Length)

- $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where σ is known.
- $Z = (\bar{X} - \mu)/\sigma$ is a pivot with a standard normal distribution.
- We like to find a and b so that

$$P(a \leq Z \leq b) = 1 - \alpha$$

  with the shortest length.
- The requirement of coverage probability gives the following form of the $1 - \alpha$ confidence interval

$$\{\mu : \bar{x} - b\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} - a\frac{\sigma}{\sqrt{n}}\}$$

- The length of the above interval is $(b - a)\sigma/\sqrt{n}$.

Example 9.3.1 (Optimizing Length), continued

- We need to find a pair of numbers a and b. so that $P(a \leq Z \leq b) = 1 - \alpha$ and minimize $b - a$.
- Minimize with respect to a and b: $b - a$, subject to

$$\int_a^b f_Z(z)dz = 1 - \alpha.$$

Example 9.3.1 (Optimizing Length),continued

- By splitting the probability α equality, we obtain that $a = Z_{\alpha/2}$ and $b = -z_{\alpha/2}$. We can find out by a numerical study that such the choice is optimal.
- The strategy of spitting α equally, which is optimal in this case, is not always optimal.
- What makes the equal α is the fact that the height of the pdf is the same at $z_{\alpha/2}$ and $z_{\alpha/2}$.
- The density function $f(.)$ is called unimodal if there exists x^* such that $f(x)$ is nondecreasing for $x \leq x^*$ and is nonincreasing for $x \geq x^*$.

Theorem 9.3.2

- Let $f(x)$ be a unimodal pdf. Le the interval $[a, b]$ satisfies the following properties: (i) $\int_a^b f(x)dx = 1 - \alpha$; (ii) $f(a) = f(b) > 0$, and (iii) $a \leq x^* \leq b$, where x^* is a mode of $f(x)$.
- Then, $[a, b]$ is the shortest among all intervals that satisfies (i).

Proof

- Let $[a', b']$ be any interval with $b' - a' < b - a$. We will show that this implies $\int_{a'}^{b'} f(x)dx < 1 - \alpha$.
- We will show this result for $a' \leq a$, and the proof being similar if $a < a'$.
- We need to consider two cases, $b' \leq a$ and $b' > a$.
- If $b' \leq a$, then, $a' \leq b' \leq a \leq x^*$ and

$$\int_{a'}^{b'} f(x)dx \leq f(b')(b' - a') \leq f(a)(b' - a')$$

$$< f(a)(b - a) \leq \int_{a}^{b} f(x)dx = 1 - \alpha,$$

completing the proof in the first case.

Proof for Case 2

- If $b' > a$, then $a' \leq a < b' < b$, which implies that

$$\int_{a'}^{b'} f(x)dx = \int_{a}^{b} f(x)dx + [\int_{a'}^{a} f(x)dx - \int_{b'}^{b} f(x)dx]$$

$$= 1 - \alpha + [\int_{a'}^{a} f(x)dx - \int_{b'}^{b} f(x)dx]$$

- Note that $\int_{a'}^{a} f(x)dx \leq f(a)(a - a')$ and $\int_{b'}^{b} f(x)dx \geq f(b)(b - b')$. Thus,

$$\int_{a'}^{a} f(x)dx - \int_{b'}^{b} f(x)dx \leq f(a)(a - a') - f(b)(b - b')$$

$$= f(a)[(b' - a') - (b - a)] < 0$$

Optimizing Expected Length

- $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where σ is unknown.
- $T = (\bar{X} - \mu)/S$ is a pivot with a t distribution with $n - 1$ degrees of freedom.
- Result: the shortest length $1 - \alpha$ shortest length $1 - \alpha$ confidence interval of the form

$$\bar{x} - b\frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} - a\frac{s}{\sqrt{n}}$$

  where $a = -t_{n-1,\alpha/2}$ and $b = t_{n-1,\alpha/2}$.
- Proof; the interval length is a function of s, with general form,

$$Length(s) = (b - a)\frac{s}{\sqrt{n}}$$

- If we consider the criterion of expected length and wanted to find a $1 - \alpha$ interval to minimize

$$E_\sigma(Length(S)) = (b - a)c(n)\frac{\sigma}{\sqrt{n}}$$

Optimizing Expected Length

- Then, by Theorem 9.3.2, we obtain that the choice $a = -t_{n-1,\alpha/2}$ and $b = t_{n-1,\alpha/2}$.

Tests versus Confidence Sets

- Confidence intervals are more informative than tests. Look at Figure 1.
- Suppose we are testing $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$.
- We see 5 different confidence intervals. The first two cases (top two) correspond to not rejecting H_0. The other three correspond to rejecting H_0.
- Reporting the confidence intervals is much more informative than simply reporting "reject" or "don't reject".

Tests versus Confidence Intervals

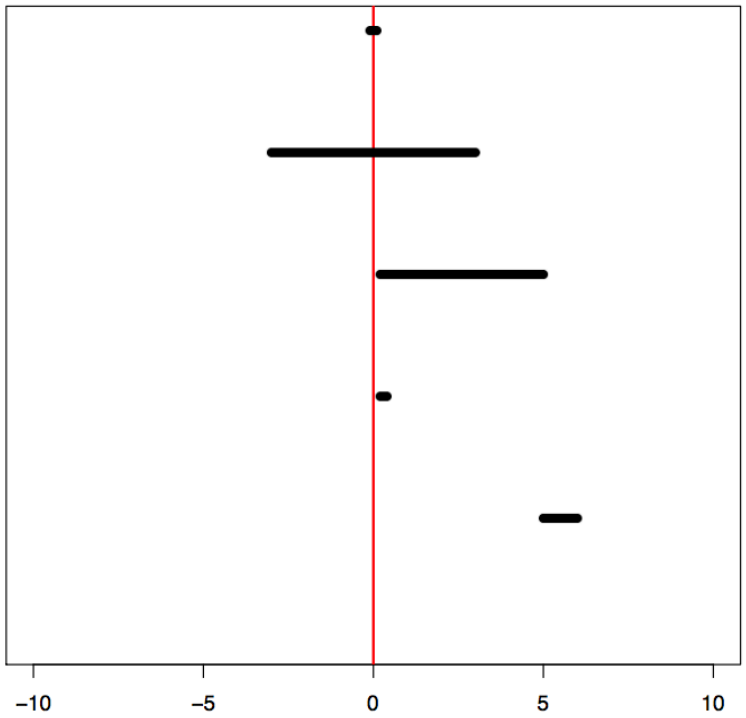


Figure 1: Five examples: 1. Not significant, precise. 2. Not significant, imprecise. 3. Barely significant, imprecise. 4. Barely significant, precise. 5. Significant and precise.